

Labor Supply Shocks with Fixed Prices

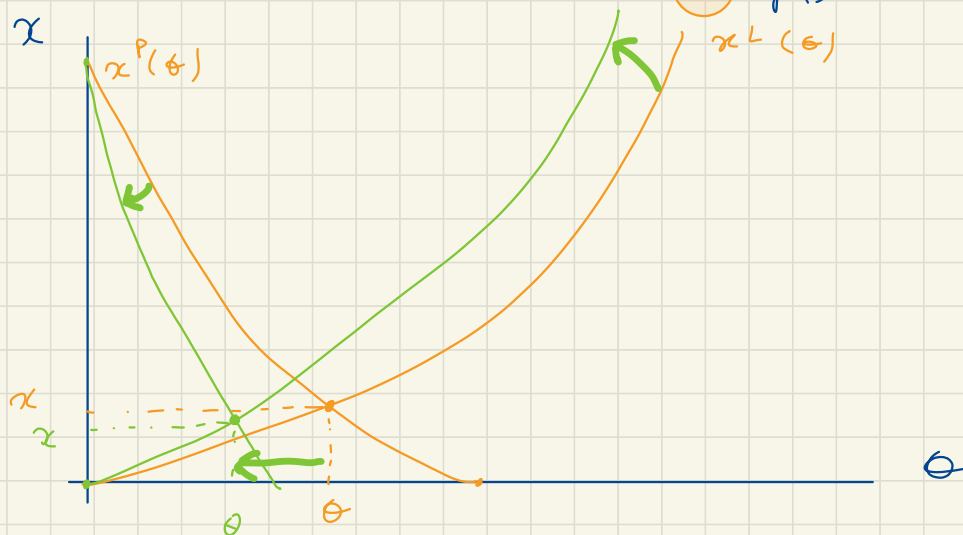
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Continue labor supply phase.

increase in h , size of labor force.

$$\cdot x = x^L(\theta) = f^{-1} \left(\frac{w/p}{a\alpha} l^{1-\alpha} \hat{f}(\theta)^{1-\alpha} [1+\tau(\theta)]^\alpha \right)$$

$$\cdot x = x^P(\theta) = \tau^{-1} \left(\left[\frac{x^\xi \mu \alpha}{w \cdot h} \frac{1}{\hat{p}(\theta)} \right]^{1/\xi-1} \right)$$



After an increase in labor force h :

$$\cdot \underline{\theta} \downarrow [\hat{f}(\theta) \downarrow, 1 - \hat{f}(\theta) \uparrow, \hat{\tau}(\theta) \downarrow, q(\theta) \uparrow]$$

Assume that $x \uparrow$

$$\cdot \gamma \downarrow \text{ b/c } \gamma = \frac{x^\xi}{[1+\tau(\theta)]^{\xi-1}} \frac{\mu}{p} \quad (AD)$$

$$\cdot \frac{w \cdot e}{p \cdot \gamma} = \alpha \quad \text{so } l \downarrow$$

$$l = \left[\frac{f(x) \alpha \omega}{w/p} \right]^{1/(1-\alpha)} \cdot \left[\frac{1}{1+\hat{\tau}(\theta)} \right]^{2/(1-\alpha)} \quad (LD)$$

so $l \uparrow$

$\Rightarrow$ contradiction: the model equations cannot all be satisfied if $\alpha \uparrow$

$\Rightarrow$ so $\alpha \downarrow$ ($f(x) \downarrow$, $1-f(x) \uparrow$, $\tau(x) \downarrow$)

$$- \quad y = \frac{x^\alpha}{(1+\tau(x))^{\alpha-1}} \frac{\nu}{p} \quad (AD) \quad \text{so } y \uparrow$$

$$- \quad \text{labour share } \frac{w \cdot l}{p \cdot y} = \alpha \quad \text{so } l \uparrow$$

$$- \quad c = y / (1+\tau(x)) \quad \text{so } c \uparrow$$

$$- \quad m = l / (1+\hat{\tau}(\theta)) \quad \text{so } m \uparrow$$

Labour supply shock is the only shock that leads to higher employment l but lower labour market tightness θ .